Modeling and Validation of a Soccer Ball's Three-Dimensional Flight to the Goal

Problem B

Abstract

The flight of a soccer ball is shaped by gravity, aerodynamic drag and the force associated with spin. A model of these effects provides a way to describe the ball's three-dimensional path from a penalty kick to the goal. We establish a system of differential equations for the ball's position, velocity and spin, and solve it numerically to locate the first crossing of the goal plane.

The task gives the ball-centre position at launch and at two later instants, 0.05 s and 0.10 s. We fit candidate flight states to those three locations and propagate them to the goal plane. In the ideal constant-acceleration baseline, the three points yield a goal-plane position of approximately $(2.495, 1.920)$ m. A bounded scan of spin, wind, density and drag keeps candidates whose predicted early positions agree with the observations within a stated tolerance. Among 600 sampled candidates, 495 are retained and reach the goal plane, with sampled coordinates $y = 2.478$–3.775 m and $z = 1.132$–2.730 m.

To test the method, we use recorded shots from two matches in the DFL tracking dataset. At 25 Hz, the available early positions are at 0, 0.04 and 0.08 s; later frames are reserved to establish the actual goal-plane crossing. We report crossing failures alongside errors. These examples motivate further tests using independent datasets.

Keywords: Soccer ball, flight trajectory, aerodynamic drag, Magnus effect, model validation

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1 Introduction

1.1 Problem Background

The path of a kicked soccer ball is not generally a straight line. Gravity bends the trajectory downwards, air resistance slows the ball, and its rotation can produce a sideways or vertical aerodynamic force. The combination of these effects determines where the ball reaches the plane of the goal. Mathematical models of soccer-ball flight have therefore been developed using Newton's law together with experimental aerodynamic relationships [1, 2, 5]. High-fidelity simulations have also shown how seams and other surface features can alter the flow and the trajectory [6].

An effective way to examine a physical model is to compare its calculated path with positions recorded during a real kick. The DFL open dataset combines match events and optical tracking data for professional matches [3, 4]. These records make it possible to identify a penalty event, follow the ball through the flight, and compare the model with its position at the actual goal plane.

1.2 Problem Restatement

We model a soccer ball travelling in three dimensions from the vicinity of the penalty mark to the goal. The goal is 7.32 m wide and 2.44 m high; the penalty mark is 11 m from the goal line. We establish a model incorporating gravity, drag and the Magnus effect, solve for its trajectory and the point at which it crosses the goal plane, and compare the calculated positions with a recorded penalty flight. The prediction uses the initial position and the positions at 0.05 and 0.10 s; subsequent positions are reserved for validation. Figure 1 presents the broader research workflow; the early positions are the model inputs and later tracked positions provide the validation target.

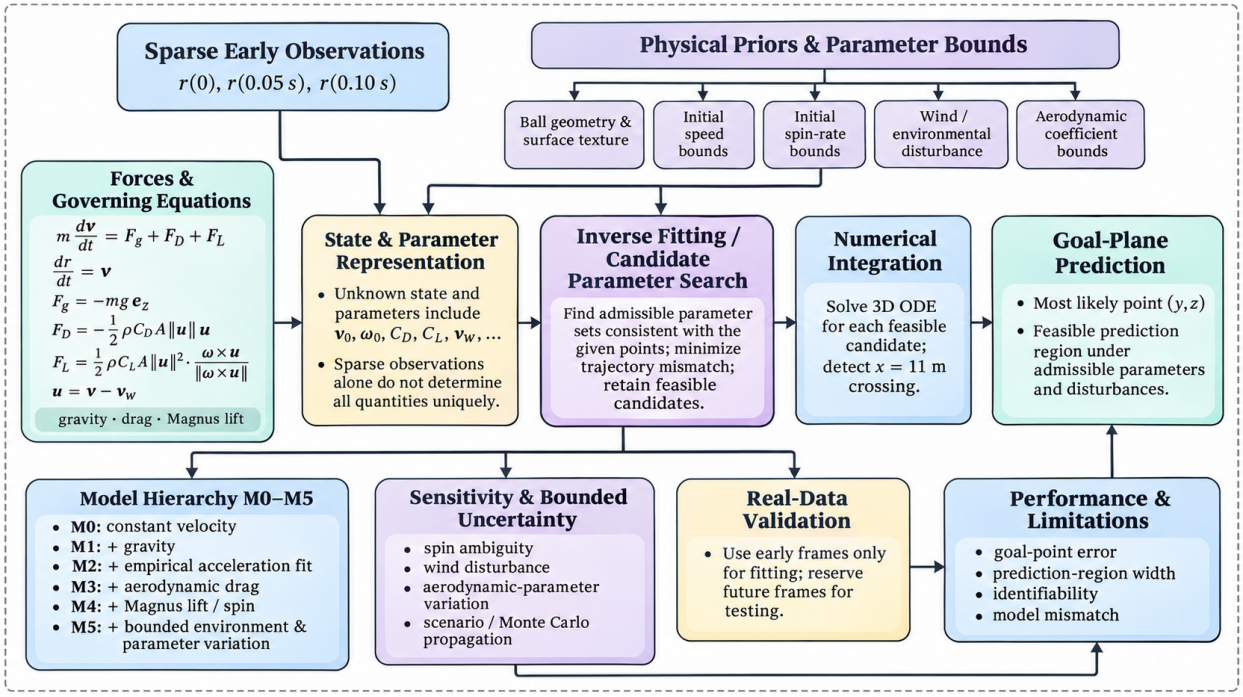


Figure 1: Overall framework for trajectory modeling, bounded uncertainty and subsequent prediction.

2 Preparation of the Models

2.1 Assumptions

- The ball is first idealized geometrically as a rigid sphere of constant mass and radius, so its cross-sectional area is πR^2 and its position represents its centre. This idealization follows the centre-of-mass approach in soccer-ball flight models [5, 2, 7]. Seams and surface texture are not explicitly resolved in the geometry; their aerodynamic influence is considered in the sensitivity analysis. The reference force coefficients are empirical soccer-ball coefficients, rather than coefficients established for a perfectly smooth sphere.

- During free flight, gravity, aerodynamic drag and spin-induced lift are the principal forces, as in the flight models of Maponi and Piombin and Bray and Kerwin [5, 2]. Free-flight integration stops at ground contact; the analysis assumes no additional contact before the recorded goal-plane crossing.

- Air density is uniform over the short flight. The reference comparison assumes still air and constant angular velocity; Maponi and Piombin describe wind and spin evolution as additional inputs [5]. These are simplifying choices for the present comparison.

- The first position and the next two available early positions are used to fit the state; later positions are retained for validation using the available DFL tracking data [3].

- The tracking data's vertical coordinate is treated as a height in metres for the present calculation. The dataset description identifies it as ball height [3]; its unit and reference are working assumptions here.

2.2 Notations

The main notations and reference values used in the model are shown in Table 1.

Table 1: Notations

Symbol	Definition	Reference value
$\boldsymbol{r} = (x, y, z)$	Ball-centre position	m
$\boldsymbol{v}$	Ball velocity	m/s
$\boldsymbol{\omega}$	Angular velocity	rad/s
m	Ball mass	0.43 kg
R	Ball radius	0.11 m
ρ	Air density	1.20 kg/m^3
g	Gravitational acceleration	9.81 m/s^2
A	Cross-sectional area, πR^2	m^2
C_D, C_L	Drag and lift coefficients	Dimensionless
$\boldsymbol{u}$	Velocity relative to air	m/s
X_g	Goal-plane coordinate	52.50 m (match coordinates)

3 Model Establishment

We express the ball's motion using one set of equations and switch individual forces on or off to obtain simpler comparison models. The coordinate x points towards the goal, y is horizontal across the goal, and z is vertical. In the selected match the coordinate origin is the centre of the pitch.

3.1 Geometry and Goal Plane

For a pitch 105 m long with its origin at the centre, the two goal lines are located at $x = \pm 52.50$ m. The selected kick travels towards the positive-x goal. For the competition geometry the valid goal opening is $-3.66 \leq y \leq 3.66$ m and $0 \leq z \leq 2.44$ m. The goal-plane position is the ball centre's first forward intersection with $x = X_g$.

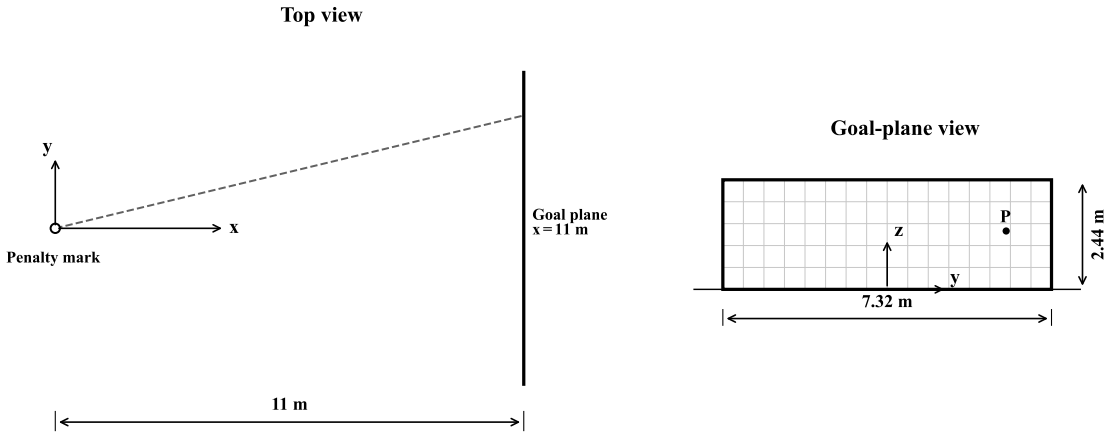


Figure 2: Penalty-kick geometry in local coordinates. The dashed line and point P indicate an illustrative crossing.

Figure 2 uses the penalty mark as the local origin and places the goal plane at $x = 11$ m. For the match record below, the pitch-centred coordinate system instead places the goal at $X_g = 52.50$ m.

3.2 State Variables and Newton's Law

The state consists of position $\boldsymbol{r}$, velocity $\boldsymbol{v}$ and spin $\boldsymbol{\omega}$. Three-dimensional models of a spinning ball have also been studied analytically and numerically by Myers and Mitchell [7]. Newton's law gives

$$\frac{d\boldsymbol{r}}{dt} = \boldsymbol{v}, \qquad m\frac{d\boldsymbol{v}}{dt} = \boldsymbol{F}_g + \boldsymbol{F}_D + \boldsymbol{F}_L. \tag{1}$$

The initial position and motion parameters define the subsequent flight.

3.3 Gravity

With the vertical unit vector $\boldsymbol{e}_z$, the gravitational force is

$$\boldsymbol{F}_g = -mg\boldsymbol{e}_z. \tag{2}$$

When drag and lift are removed, Eq. (1) reduces to the familiar parabolic trajectory.

3.4 Aerodynamic Drag

Let $\boldsymbol{u} = \boldsymbol{v} - \boldsymbol{v}_{\text{wind}}$ be the velocity relative to the air. Drag acts against this motion:

$$\boldsymbol{F}_D = -\tfrac{1}{2}\rho C_D A\|\boldsymbol{u}\|\boldsymbol{u}. \tag{3}$$

The drag coefficient depends on the ball and its flow conditions; it is prescribed or estimated within a chosen range.

3.5 Spin-Induced Lift

The direction of the Magnus force is given by $\boldsymbol{\omega} \times \boldsymbol{u}$. Let θ be the angle between the spin axis and relative airflow. In our three-dimensional extension of the transverse-spin relationship, the force is

$$\boldsymbol{F}_L = \tfrac{1}{2}\rho C_L A \|\boldsymbol{u}\|^2 \sin\theta \, \frac{\boldsymbol{\omega} \times \boldsymbol{u}}{\|\boldsymbol{\omega} \times \boldsymbol{u}\|}. \tag{4}$$

The force is set to zero when the cross product vanishes. The factor $\sin\theta$ smoothly reduces lift as the spin axis approaches the airflow direction; this is a modeling extension of the transverse-spin relation.

3.6 Aerodynamic Coefficients

Linthorne and Patel use the following relationships for a spinning soccer ball [1]:

$$C_L = 0.25\left(1 - e^{-0.5f}\right), \qquad C_D = C_{D0} + 0.014f, \tag{5}$$

where $f = \|\boldsymbol{\omega}\|/(2\pi)$ is the spin rate in revolutions per second. The paper discusses C_{D0} of approximately 0.1–0.2 for speeds near 25–35 m/s. The original work considers punt kicks; applying the relation to a three-dimensional penalty shot is the modeling choice examined here.

3.7 Spin and Environmental Inputs

The optional spin-decay equation is

$$\frac{d\boldsymbol{\omega}}{dt} = -k_\omega \boldsymbol{\omega}. \tag{6}$$

For the reference fit, $k_\omega = 0$ and $\boldsymbol{v}_{\text{wind}} = \mathbf{0}$. Both quantities can be changed for other experiments.

3.8 Numerical Solution and Comparison Models

Equation (1) is integrated using an adaptive Runge–Kutta solver. Continuous event detection locates the first forward crossing of the goal plane and descending ground contact. We compare six configurations: M0 has uniform velocity; M1 adds gravity; M2 is a separate constant-acceleration fit; M3 adds drag; M4 adds constant-coefficient lift; and M5 uses the spin-dependent coefficient relationships. For M0, M1, M3, M4 and M5 only the initial velocity is fitted in the fixed-parameter comparison. A second M5 fit also adjusts spin and C_{D0}.

4 Model Solving

For the competition example, the three input observations are

$$\boldsymbol{r}(0) = (0, 0, 0.11), \quad \boldsymbol{r}(0.05) = (1.50, 0.45, 0.44), \quad \boldsymbol{r}(0.10) = (2.96, 0.86, 0.74)\ \text{m}. \tag{7}$$

Each model uses the available points to estimate its unknown state, then predicts the first crossing of $x = 11$ m. A three-point constant-acceleration fit gives $(y, z) = (2.495, 1.920)$ m. A

fixed-parameter gravity-and-drag fit, using all three positions and zero spin, gives $(3.217, 1.907)$ m. These are model-dependent estimates for the stated input.

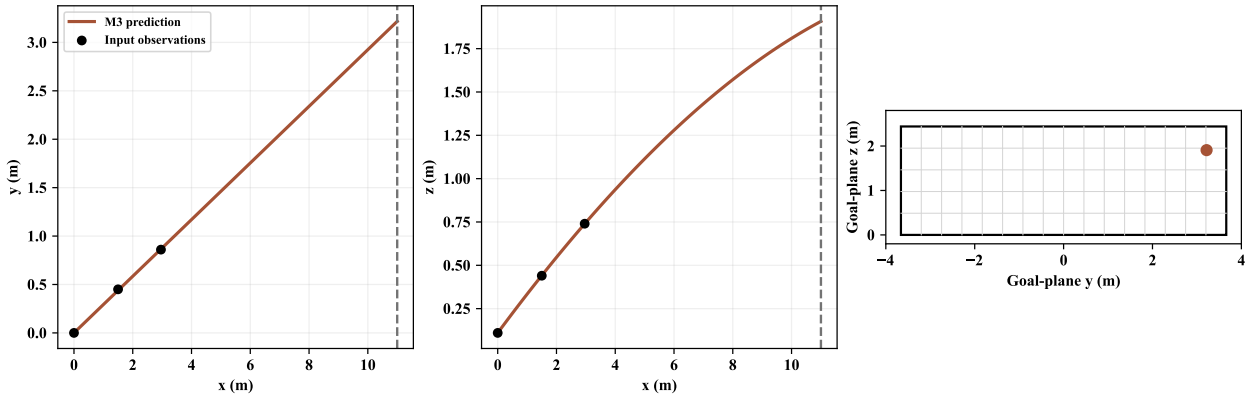


Figure 3: Three input positions and the gravity-and-drag model: top view, side view, and predicted goal-plane crossing.

For recorded shots, the tracking rate is 25 Hz, so we use the three actually available positions at 0, 0.04 and 0.08 s. Later frames are used only to interpolate the true crossing of the actual goal-line plane. Two DFL matches contribute four selected events under the documented common criteria; the two matches are part of the same dataset. Recorded height is treated as metres. A second dataset, ball3d, could not be accessed without a Google sign-in in this session.

All four examples are tested under the same three-frame input. M2 predicts a plane crossing in two of four cases; fixed-spin M5 reaches it in two cases, while M5 with a spin prior reaches it in three. A single event is common to all models: its goal-plane errors are 13.528 m for M2, 3.220 m for fixed-spin M5, and 3.308 m for M5 with a spin prior. Unreached cases are recorded separately; means computed from different success subsets are not directly comparable. The common event is a header in the second match, and the association between event and tracking timestamps requires further verification.

Previously, M2 and M5 were fitted to complete recorded trajectories. That experiment described reconstruction rather than an early prediction. M2's greater flexibility and its use of the later positions can make such a fit look favourable. The present test retains those later positions solely for the goal-plane answer.

5 Sensitivity Analysis

We vary one parameter at a time around a reference M5 state fitted to the three competition observations, while retaining its initial position and the remaining parameters. Each resulting trajectory is integrated to the same goal plane. The purpose is to see how the calculated crossing moves when a physical input changes.

5.1 Launch Speed

Multiplying the fitted velocity by 0.70 to 1.30 gives sampled goal-plane heights from 1.148 m to 2.232 m. The faster ball spends less time in flight and reaches the plane at a greater height.

5.2 Horizontal Launch Angle

Rotating the horizontal velocity from -10° to $+10^\circ$ gives sampled crossing coordinates from $y = 1.789$ m to $y = 6.292$ m. Direction strongly affects the horizontal location in this experiment.

5.3 Spin

Scaling all three fitted spin components from zero to four times their reference values gives sampled crossing ranges $y = 3.014$–3.969 m and $z = 1.857$–1.931 m. This isolates a change in spin magnitude along the fitted axis.

5.4 Drag Coefficient

Increasing C_{D0} from 0.05 to 0.40 changes the calculated height across 1.848–1.968 m for the chosen reference state. The extended range is a scenario scan, wider than the 0.1–0.2 range discussed for the reference model.

Surface and coefficient-law comparison. A real soccer ball has panels and seams, whereas the idealized sphere has no resolved surface features. Paudel and Yan compared a geometrically detailed soccer ball with a smooth sphere at the same mass and moment of inertia; their CFD cases showed stronger lateral Magnus deflection for the smooth sphere [6]. Maponi and Piombin tabulated speed-dependent drag coefficients for a soccer-ball model [5].

To explore how a change of aerodynamic description affects our calculation, we run two otherwise identical *nonspinning* flights from the fitted reference position and velocity. One uses the smooth-sphere $C_D(Re)$ relation reproduced by Paudel and Yan; the other uses linear interpolation of the soccer-ball drag table of Maponi and Piombin. The same mass, radius, air properties and initial conditions are used in both calculations. At the initial speed of 32.27 m/s, the respective drag coefficients are 0.086 and 0.192. The computed goal-plane heights are 1.956 m and 1.925 m, a difference of 0.031 m; their crossing times are 0.373 s and 0.386 s. Because the two coefficient laws come from different studies, this is a comparison of published drag descriptions rather than a matched reproduction of their surface CFD study. Spin is set to zero in these two runs, so this calculation isolates the drag-law contribution and does not measure the surface effect on Magnus lift.

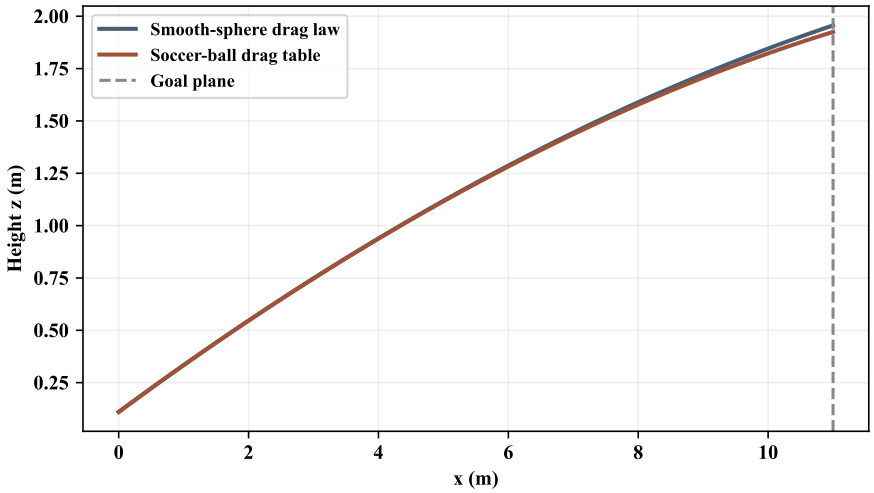


Figure 4: Nonspinning trajectories with two published drag descriptions and otherwise identical initial conditions.

5.5 Summary of Sensitivity Analysis

The selected horizontal-angle range produces the largest lateral change among the four illustrated variations, while launch speed, spin and drag also change the height. The comparison uses different chosen ranges for different quantities. The coefficient-law comparison further illustrates how the aerodynamic description affects flight time and height. The computed cases and their assumptions are saved with the project.

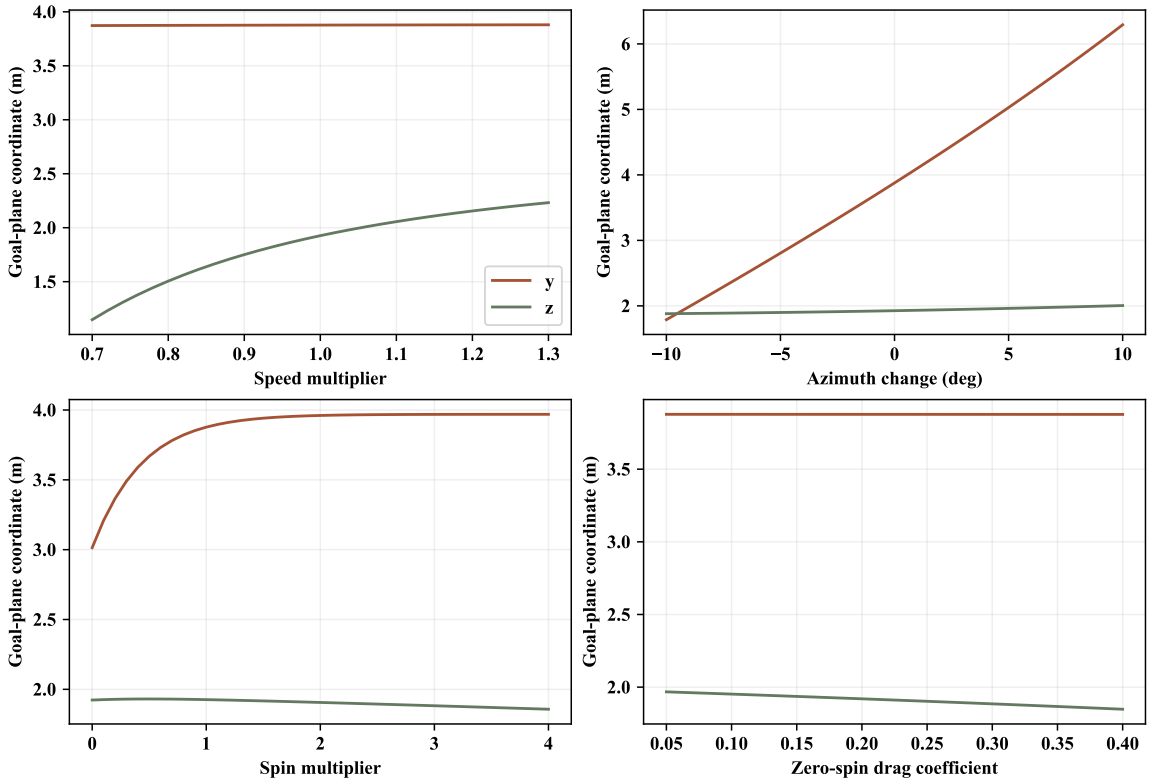


Figure 5: One-factor changes in goal-plane coordinates around the fitted reference shot.

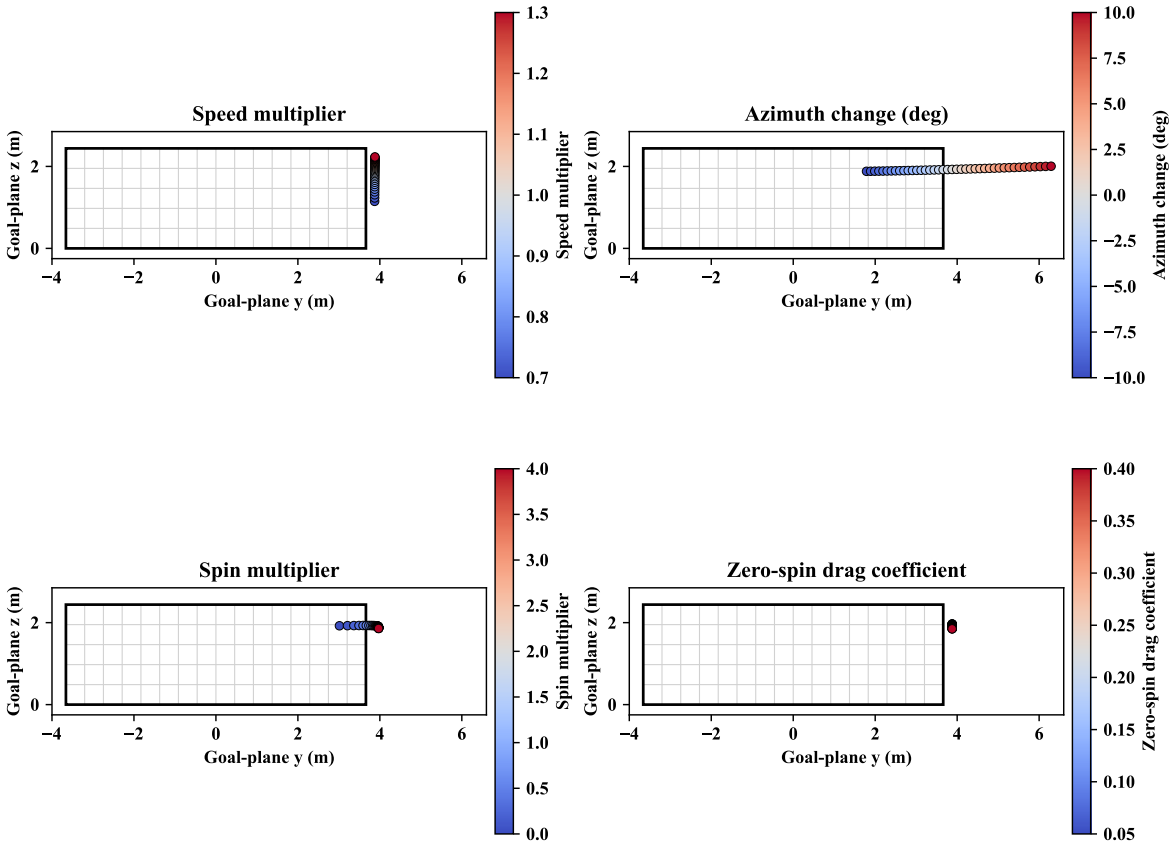


Figure 6: Goal-plane positions for the four one-factor scans. Blue and red indicate lower and higher values of the varied parameter; positions outside the goal frame are also shown.

5.6 Bounded Uncertainty

We distinguish observation, parameter, environmental and model uncertainty. Here the three given positions are treated as exact inputs. Initial spin is unknown: each component is sampled within $[-50, 50]$ rad/s. Each wind component ranges from -3 to 3 m/s, air density from 1.1 to 1.3 kg/m^3, and C_{D0} from 0.1 to 0.2. These bounds are explicit scenario choices. For each sampled combination, the initial velocity is fitted using only the three positions. Candidates with a maximum component residual greater than 0.02 m are rejected; this is a compatibility tolerance for numerical model mismatch, not an assumed camera-error distribution.

With seed 20261107, 600 combinations were drawn, 105 failed the compatibility criterion and 495 reached the goal plane. The retained sampled crossings range from $y = 2.478$ to 3.775 m and $z = 1.132$ to 2.730 m. Figure 7 shows the sampled region. It is a finite-sample range for the specified bounds and tolerance, rather than a guaranteed enclosure or probability interval. The comparison of alternative coefficient laws illustrates model-form uncertainty.

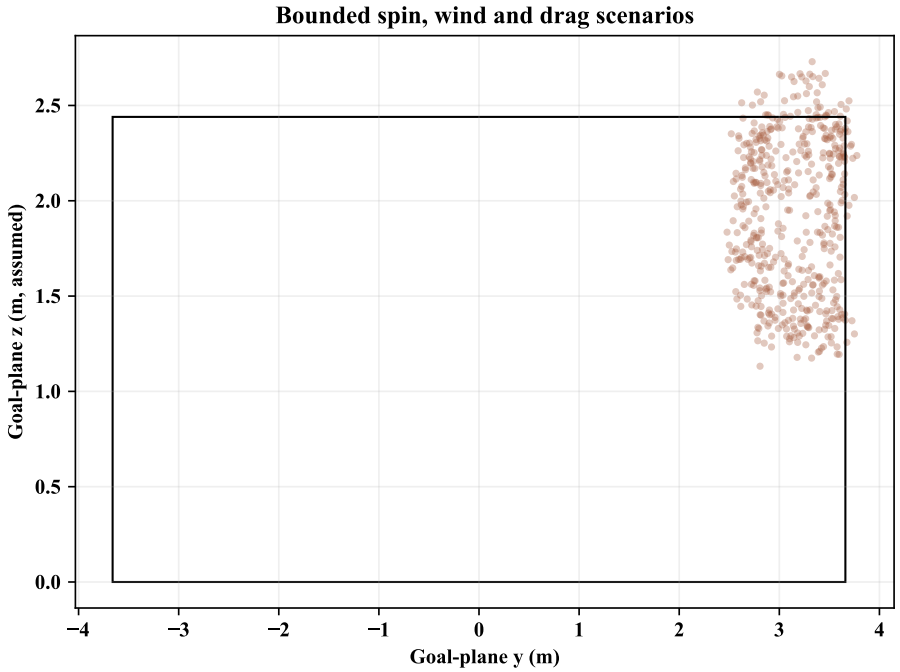


Figure 7: Sampled goal-plane positions compatible with the three early observations.

6 Conclusion

We developed a three-dimensional force model and used three early positions to estimate the state and predict the goal-plane crossing. For the competition example, a constant-acceleration baseline gives $(2.495, 1.920)$ m, while the fixed gravity-and-drag model gives $(3.217, 1.907)$ m. Bounded scans show how assumed spin, wind and drag affect the possible crossing region. The four recorded events from two matches illustrate both the method and cases where a model does not reach the goal plane. More independent data and better characterization of spin and tracking errors will improve evaluation.

7 Model Evaluation

7.1 Model Strengths

- The model combines recognizable physical forces in a single set of equations, while allowing individual effects to be switched on and off.
- Event detection determines the goal-plane intersection and identifies ground contact during numerical integration.
- The validation uses three early tracking positions as inputs and later recorded positions to establish the actual goal-plane crossing.

7.2 Model Weaknesses

- The vertical tracking coordinate is treated as metres pending confirmation of its unit and reference.
- The early-prediction comparison uses four shots from two matches of one dataset; precise kick or header times within the initial frame interval are not established.
- Spin is not independently recorded and several parameter combinations can fit a small number of early positions.

7.3 Model Improvements

Further work can check the tracking system's height definition and align the kick time using synchronized video or higher-rate measurements. A useful next step is to compare seam-resolving and smooth-sphere aerodynamic coefficients under matched speed, spin and air conditions, following the physical comparison of Paudel and Yan [6]. Additional penalty trajectories, together with measured ball spin and environmental conditions, would support a broader comparison of the force models. Additional independent trajectories and measurements of spin can improve validation of early goal-plane predictions.

References

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Python Code

The script below performs the bounded parameter scan. The physics and plotting code is available in the project repository.

```
"""Sample explicitly bounded spin, wind and drag and record all event outcomes."""
import csv,json
from dataclasses import replace
from pathlib import Path
import numpy as np
from football.physics import PhysicsConfig,simulate,linthorne_patel_coefficients
from football.inference.physical_fit import fit_velocity
rng=np.random.default_rng(20261107); n=600
t=np.array([0.,.05,.10]); obs=np.array([[0,0,.11],[1.50,.45,.44],[2.96,.86,.74]]);p=obs[0]
out=Path('results/bounded_uncertainty');out.mkdir(parents=True,exist_ok=True);rows=[]
for i in range(n):
    spin=rng.uniform(-50,50,3);wind=rng.uniform(-3,3,3);rho=rng.uniform(1.1,1.3);cd=rng.uniform(.1,.2)
    cfg=replace(PhysicsConfig.for_model('M5'),air_density=rho,wind=tuple(wind),coefficients=linthorne_patel_coefficients(cd0=cd))
    fit=fit_velocity(t,obs,cfg,sigma=.01,spin=spin);v=fit['velocity'];misfit=float(np.max(np.abs(fit['residuals'][1:])))
    if misfit>.02:
       rows.append({'i':i,'spin_x':spin[0],'spin_y':spin[1],'spin_z':spin[2],'wind_x':wind[0],'wind_y':wind[1],'wind_z':wind[2],'rho':rho,'cd0':cd,'event':'incompatible','y':'','z':''});continue
    f=simulate(p,v,spin=spin,config=cfg,t_span=(0,2),plane_x=11.)
    point=f.event['position'] if f.event['type']=='plane' else None
    rows.append({'i':i,'spin_x':spin[0],'spin_y':spin[1],'spin_z':spin[2],'wind_x':wind[0],'wind_y':wind[1],'wind_z':wind[2],'rho':rho,'cd0':cd,'event':f.event['type'],'y':point[1] if point is not None else '', 'z':point[2] if point is not None else ''})
with (out/'samples.csv').open('w',encoding='utf8',newline='') as f:w=csv.DictWriter(f,fieldnames=rows[0]);w.writeheader();w.writerows(rows)
valid=[r for r in rows if r['event']=='plane']
stats={'seed':20261107,'samples':n,'input_times_s':t.tolist(),'positions_m':obs.tolist(),'max_component_fit_tolerance_m':.02,'spin_bounds_rad_s':[-50,50],'wind_component_bounds_m_s':[-3,3],'rho_kg_m3':[1.1,1.3],'cd0':[.1,.2], 'plane':len(valid),'ground':sum(r['event']=='ground' for r in rows),'none':sum(r['event']=='none' for r in rows),'incompatible':sum(r['event']=='incompatible' for r in rows),'y_range':[min(r['y'] for r in valid),max(r['y'] for r in valid)],'z_range':[min(r['z'] for r in valid),max(r['z'] for r in valid)]}
(out/'summary.json').write_text(json.dumps(stats,indent=2),encoding='utf8');print(json.dumps(stats,indent=2))
```